VistaLux 2025 Growth Projections

Our approach

- Our current model predicts quarterly growth from four economic indicators.
- We tested two alternatives to see if they could project better.
- They help a little, and all three projections tell the same story.

Our current model

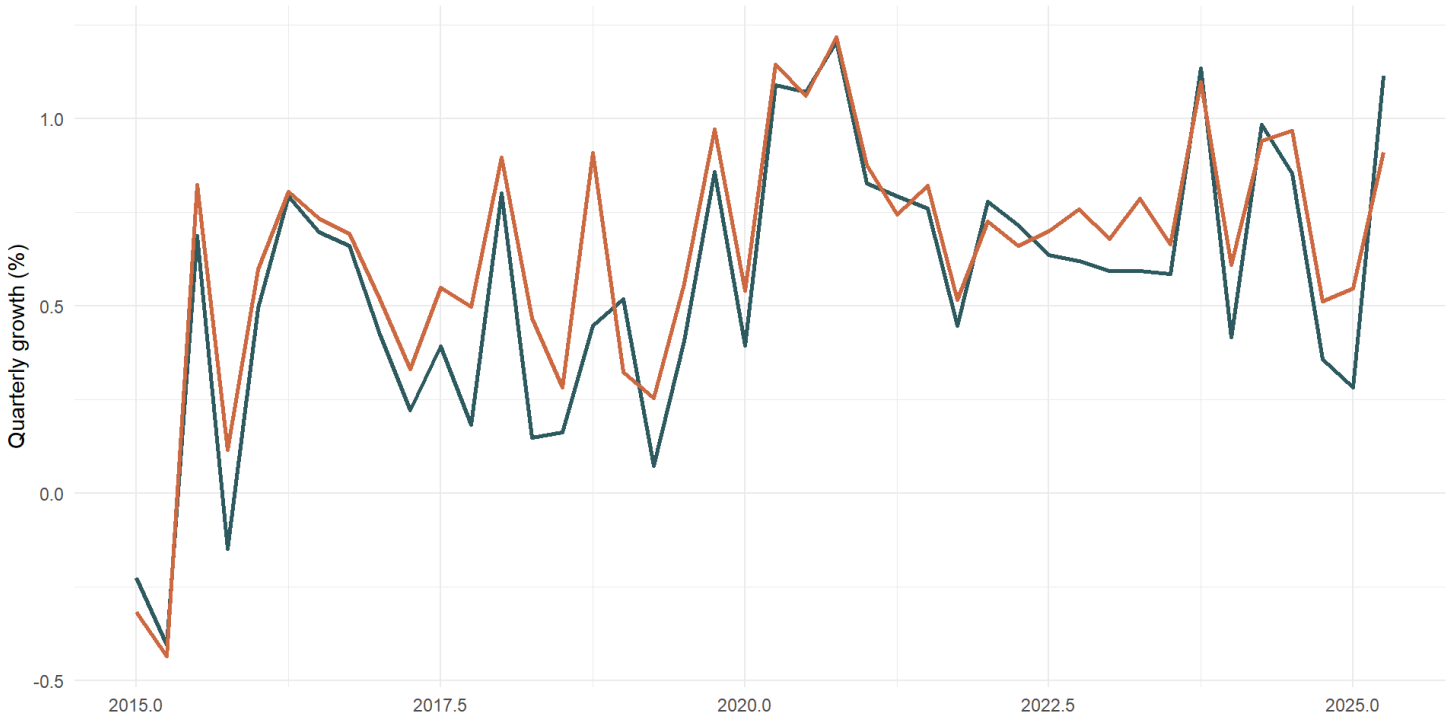
$$\text{Growth} = \beta_0 + \beta_1 \text{Income} + \beta_2 \text{Production}$$

$$+ \beta_3 \text{Savings} + \beta_4 \text{Unemployment} + \varepsilon_i$$

$$\varepsilon_i \sim \text{iid } N(0, \sigma^2)$$

- The model is one line per quarter, four indicators in and growth out.
- It is fit on 198 quarters from 1976 to mid-2025 and explains about 77% of the variation in growth.
- To project a quarter, we plug that quarter's indicator values into the equation.

How the current model performs



- The fitted line tracks actual growth closely over the last decade.
- Its predictions were off by about 0.19 points per quarter on the last four quarters.
- Under this model, Q3 projects to 0.81% and Q4 to 0.70%.
- It cannot project a quarter unless the indicator values already exist.

Why we tested alternatives

Check	p-value
Constant variance	0.024
Normal errors	0.044
Independent quarters	0.029

- Every check we ran on the model's errors came back showing a problem.
- The big one is that quarters are correlated, and that correlation is what a time series model can take advantage of.

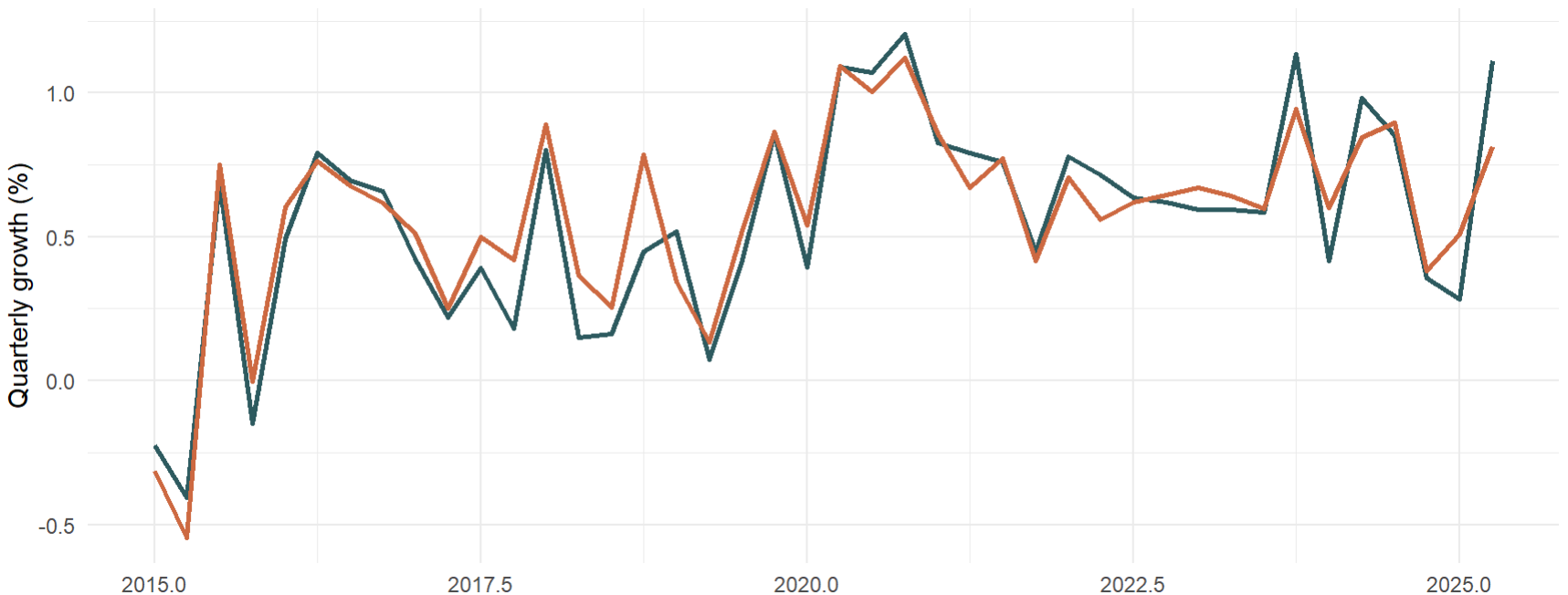
Alternative 1, regression with time-aware errors

- This keeps the same equation but lets the errors follow a seasonal ARIMA pattern.
- A bad quarter now carries information into the next projection.
- auto.arima selected an ARIMA(2,1,1)(1,0,1)[4] structure for the errors.

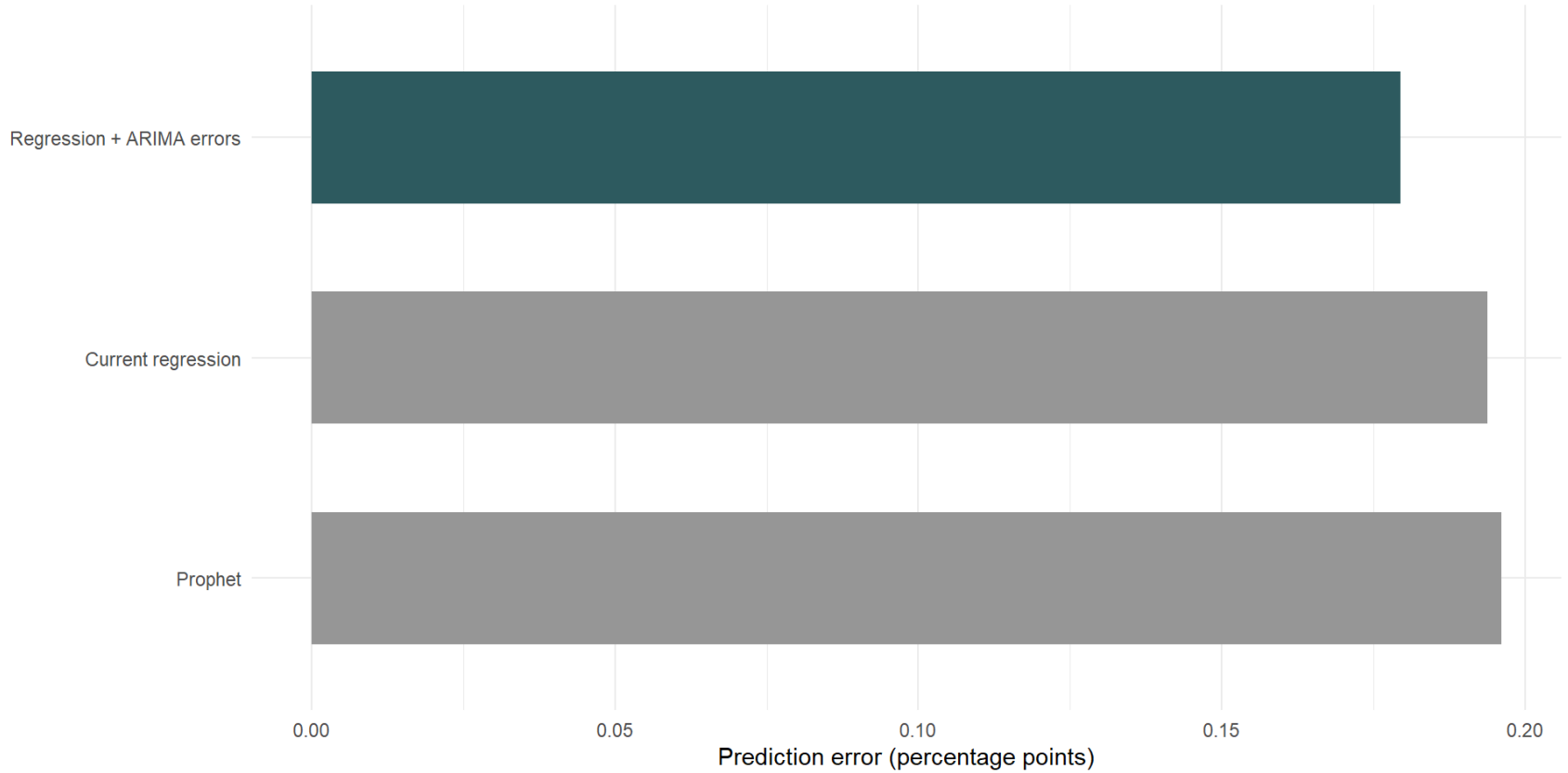
```
        ar1          ar2          ma1         sar1         sma1       Income
-0.10901747   0.10984393  -0.97529731  -0.82925030   0.78135733   0.73619473
 Production      Savings Unemployment
 0.03606896  -0.05220315  -0.23142804
```

Alternative 2, Prophet with the same indicators

- Prophet draws a smooth trend plus a seasonal curve through the growth history.
- We feed it the same four indicators as extra inputs.
- It is built for forecasting, so it handles the calendar directly.



Head to head, which predicts best?



- On the last four quarters the errors were 0.18 for the ARIMA model, 0.19 for the current model, and 0.20 for Prophet.
- It is a tight race, but the time-aware errors come out ahead.

How we get Q3 and Q4 projections

Quarter	Income	Production	Savings	Unemployment
Q3	0.51	0.35	-2.61	-0.14
Q4	0.68	0.45	1.80	-0.08

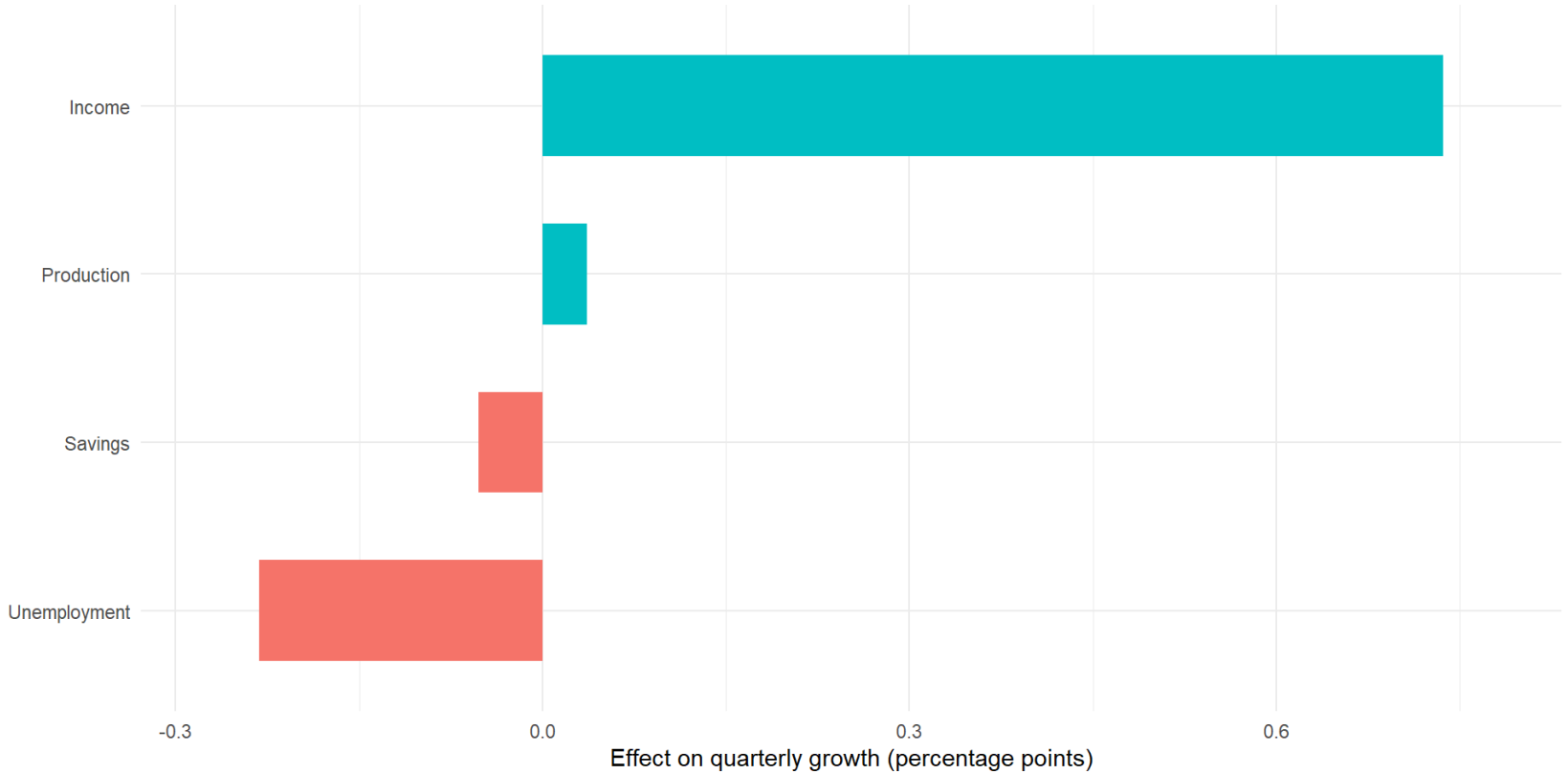
- The Q3 and Q4 indicator values do not exist yet, so we forecast them too.
- Each indicator gets its own Prophet model, projected two quarters out.
- Every model then uses the same projected inputs.

What the rest of 2025 looks like

Quarter	Current regression	Regression + ARIMA	Prophet
Q3	0.81%	0.70%	0.73%
Q4	0.70%	0.69%	0.57%

- All three models project growth between 0.57% and 0.81% per quarter.
- The preferred model says about 0.7% in both quarters.

What actually drives VistaLux growth



- Income dominates, with one extra point of income growth worth about 0.74 points for us.
- Unemployment and savings pull growth down, while production helps a little.
- We are a discretionary-goods company, so this pattern tracks.

Recommendation

- We recommend the regression with ARIMA errors because it did the best job predicting the recent quarters.
- The rest of 2025 should see steady growth around 0.7% a quarter.
- Income is the number to watch going forward.